



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

School of Computer Engineering

Information Security Lab - Project Synopsis

Homomorphic Encrypted Data Analytics Platform

Description:

Traditional data analytics requires servers to access unencrypted data, creating significant privacy risks. This project aims to develop a platform for privacy-preserving computation that performs statistical analysis (e.g., summation, averaging) directly on homomorphically encrypted data. Users will upload datasets encrypted with a scheme like Paillier, and the platform will compute and return an encrypted result. The user can then decrypt the final result locally, ensuring that the raw data is never exposed to the server, thereby guaranteeing data confidentiality throughout the analytical process.

Implementation Outline:

Client-Side Setup: Develop a user interface (using Streamlit/Dash) for generating a public/private key pair (Paillier) and encrypting a numerical dataset (e.g., a list of numbers) with the public key.

Secure Data Upload: Implement a mechanism for the client to transmit the encrypted dataset to a Python-based backend server.

Backend Computation: On the server, perform homomorphic operations (e.g., encrypted addition) on the received ciphertext to compute the desired statistic (sum, average) without decryption.

Result Transmission: Send the encrypted result from the server back to the client application.

Local Decryption & Display: The client uses their private key to decrypt the final result and displays the plaintext answer in the user interface.

Technologies & Concepts:

Cryptography: Homomorphic Encryption, Paillier Cryptosystem, Asymmetric Key Cryptography.

Backend: Python, Flask, python-paillier library for cryptographic operations.

Frontend: Streamlit or Dash for creating an interactive web-based user interface.

Concepts: Privacy-Preserving Computation, Client-Server Architecture, Secure Data Processing.

Expected Outcome:

A functioning prototype platform capable of performing basic statistical analysis on encrypted data, offering a powerful demonstration of privacy-preserving computation where data confidentiality is maintained end-to-end.

Group Member Details:

1	Aryan Dias	230911288
2	Aryaman Sonawane	230911340
3	Pranav Prabhu	230911414